

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

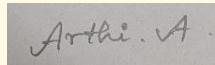
CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	ARTHI A	Reg. No.:	192525152
Date:	24.7.2026	Duration:	60 minutes

Code of Conduct

I ARTHI A (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____



Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

- 1. Understand the Problem Statement**
 - Explain the problem in your own words.
 - Identify the objectives of the proposed system.
 - State the scope of the problem.
- 2. Identify Key Stakeholders and Challenges**
 - Identify all the stakeholders involved in the system.
 - Explain the roles and responsibilities of each stakeholder.
 - Discuss the major challenges and issues faced by the stakeholders.
- 3. Analyze the Current Approach**
 - Describe the existing system or current process.
 - Identify its strengths and limitations.
 - Analyze the problems or gaps that need to be addressed.

4. Propose a Suitable Solution Using Software Engineering Principles

- Design a software-based solution for the given problem.
- Describe the major functional modules of the proposed system.
- Identify the functional and non-functional requirements.
- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. Discuss Expected Outcomes and Limitations

- Describe the expected benefits of the proposed solution.
- Explain how the solution addresses the identified challenges.
- Discuss potential risks, implementation challenges, and limitations.
- Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.

Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding	/5
Stakeholder & Challenge Analysis	/5
Current System Analysis	/5
Proposed Software Solution & Software Engineering Principles	/5
Methodology/ Framework Justification	/5
TOTAL	/5



SOFTWARE ENGINEERING CASE STUDY REPORT

College Name: SIMATS Engineering

Department: Bachelor of Technology in Artificial intelligence & Machine
learning

Course Name: Software Engineering

Assignment Title: Case Study Analysis

Case Study Title: SDG 9 Smart Construction Project Management System

Student Name: Arthi A

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1. Introduction

Construction companies today handle several projects in parallel, and most still rely on spreadsheets, informal messaging, and manual site reports to track progress. This fragmented approach leads to serious problems: delayed material procurement because purchase requests get lost in communication chains, miscommunication between site engineers and the head office, and no real-time visibility into how far a project has actually progressed. The result is frequent cost overruns, missed deadlines, and disputes with clients over delivery timelines.

Solving this problem matters because the construction sector is a major contributor to national infrastructure and economic growth, yet it remains one of the least digitized industries. A centralized software solution can bring transparency, reduce waste, and help companies deliver projects on time and within budget.

This case study aligns with SDG 9: Industry, Innovation and Infrastructure, which calls for building resilient infrastructure, promoting inclusive industrialization, and fostering innovation. A digital project management system directly supports this goal by modernizing how infrastructure projects are planned, resourced, and delivered.

Objectives of the case study:

- Analyze the current manual project management process used by construction firms
- Identify gaps causing delays, miscommunication, and cost overruns
- Propose a centralized software solution covering project planning, resource management, and progress monitoring
- Recommend a suitable SDLC methodology for development
- Evaluate the expected business, social, and environmental impact of the solution

2. Problem Understanding

2.1 Problem Description

Construction companies currently manage multiple ongoing projects using disconnected tools — spreadsheets for budgeting, phone calls or messaging apps for site updates, and paper-based logs for material tracking. Because these tools don't communicate with each other, project managers often don't know the real status of a site until a scheduled review meeting, by which point delays have already occurred. Material procurement is especially affected: site engineers request materials informally, requests get lost or delayed, and work stalls while waiting for supplies that were ordered too late. There is no single source of truth showing, at any moment, what stage each project is at, what resources have been used, and what is still pending.

2.2 Objectives

- Reduce manual, paper-based and spreadsheet-based work
- Improve real-time visibility into project progress
- Increase transparency between site teams, procurement, and management
- Enable faster and more accurate material procurement
- Improve resource allocation across multiple simultaneous projects
- Reduce cost overruns caused by delays and miscommunication
- Enhance accountability through role-based access and audit trails
- Improve overall client satisfaction through timely delivery

2.3 Scope

In Scope:

- Project planning and scheduling module
- Resource (labor, material, equipment) management
- Real-time progress tracking and reporting
- Procurement request and approval workflow
- Notifications and alerts for delays or shortages
- Role-based dashboards for admins, site engineers, and clients

Out of Scope:

- Financial accounting / payroll systems (assumed to integrate with existing ERP, not rebuilt)
- Heavy equipment IoT sensor integration (listed under future enhancements)

3. Stakeholder Analysis

Stakeholder	Role
Project Manager (Administrator)	Oversees all projects, approves budgets and schedules, monitors overall progress
Site Engineer	Updates daily progress, raises material and resource requests
Procurement Officer	Manages material orders, tracks vendor deliveries
Client	Views project progress, receives status reports and delay notifications
Vendor / Supplier	Receives procurement orders, updates delivery status

Stakeholder	Role
Laborer / Contractor	Logs attendance and task completion (via site engineer or mobi app)
Technician / IT Support	Maintains the system, handles technical issues and updates
Government / Regulatory Body	Reviews compliance and safety reporting where applicable

4. Existing System Analysis

4.1 Existing Process

Currently, each project is tracked in a separate spreadsheet maintained by the site engineer. Progress updates are communicated verbally or through informal messaging apps to the project manager. Material requirements are identified on-site and communicated via phone calls or handwritten notes to the procurement team, who then manually place orders with vendors. Budgets are tracked separately in accounting spreadsheets, disconnected from actual site progress data.

4.2 Strengths

- Simple to use, no training required, low upfront cost
- Flexible — engineers can adapt formats as needed
- No dependency on internet connectivity or new infrastructure

4.3 Weaknesses

- No real-time visibility into project status
- High risk of data loss, duplication, or version conflicts across spreadsheets
- Manual procurement leads to delays and stock-outs
- No centralized communication channel between stakeholders
- Difficult to consolidate reports across multiple projects
- No audit trail for approvals or changes

4.4 Gap Analysis

The current process has no mechanism for real-time monitoring, automated alerts, or centralized data storage. Delays in one project are not visible to management until significant time has passed. There is no structured workflow connecting site progress to procurement, meaning material shortages are discovered reactively rather than predicted proactively. These gaps directly cause the cost overruns and delays described in the case study.

5. Proposed Software Solution

5.1 Proposed System

The proposed solution is a centralized, web and mobile-based Construction Project Management System (CPMS) that connects project planning, resource management, and real-time progress monitoring into a single platform. Site engineers update progress directly from the field using a mobile app; the system automatically flags material needs based on project schedules; procurement officers manage vendor orders through the same platform; and project managers get a live dashboard showing the status of every project, resource utilization, and potential delays before they escalate.

5.2 Functional Modules

- User Registration & Role Management
- Login & Authentication
- Project Planning & Scheduling
- Resource (Labor, Material, Equipment) Management
- Procurement & Vendor Management
- Progress Tracking Dashboard
- Reports & Analytics
- Notifications & Alerts
- Client Portal
- Admin Panel

5.3 Functional Requirements

ID	Requirement
FR1	The system shall allow users to register and log in based on role (admin, site engineer, procurement officer, client).
FR2	The system shall allow project managers to create and schedule new projects with milestones.
FR3	The system shall allow site engineers to update daily progress from a mobile device.
FR4	The system shall allow procurement officers to raise, approve, and track material orders.
FR5	The system shall generate real-time progress reports for each project.

ID	Requirement
FR6	The system shall send automated notifications for delays, shortages, or pending approvals.
FR7	The system shall allow resource allocation (labor, equipment, material) across multiple projects.
FR8	The system shall provide clients with a read-only dashboard to view project status.
FR9	The system shall maintain an audit trail of all approvals and changes.
FR10	The system shall allow the admin to generate consolidated reports across all ongoing projects.

5.4 Non-Functional Requirements

- Performance: Dashboard updates should reflect within 5 seconds of a status change
- Security: Role-based access control and encrypted data storage
- Scalability: Should support growth from a handful of projects to hundreds
- Maintainability: Modular architecture for easy updates
- Availability: 99.5% uptime target
- Usability: Simple mobile interface usable by site staff with minimal training
- Reliability: Automated backups to prevent data loss
- Portability: Accessible on both Android and iOS, and via web browser

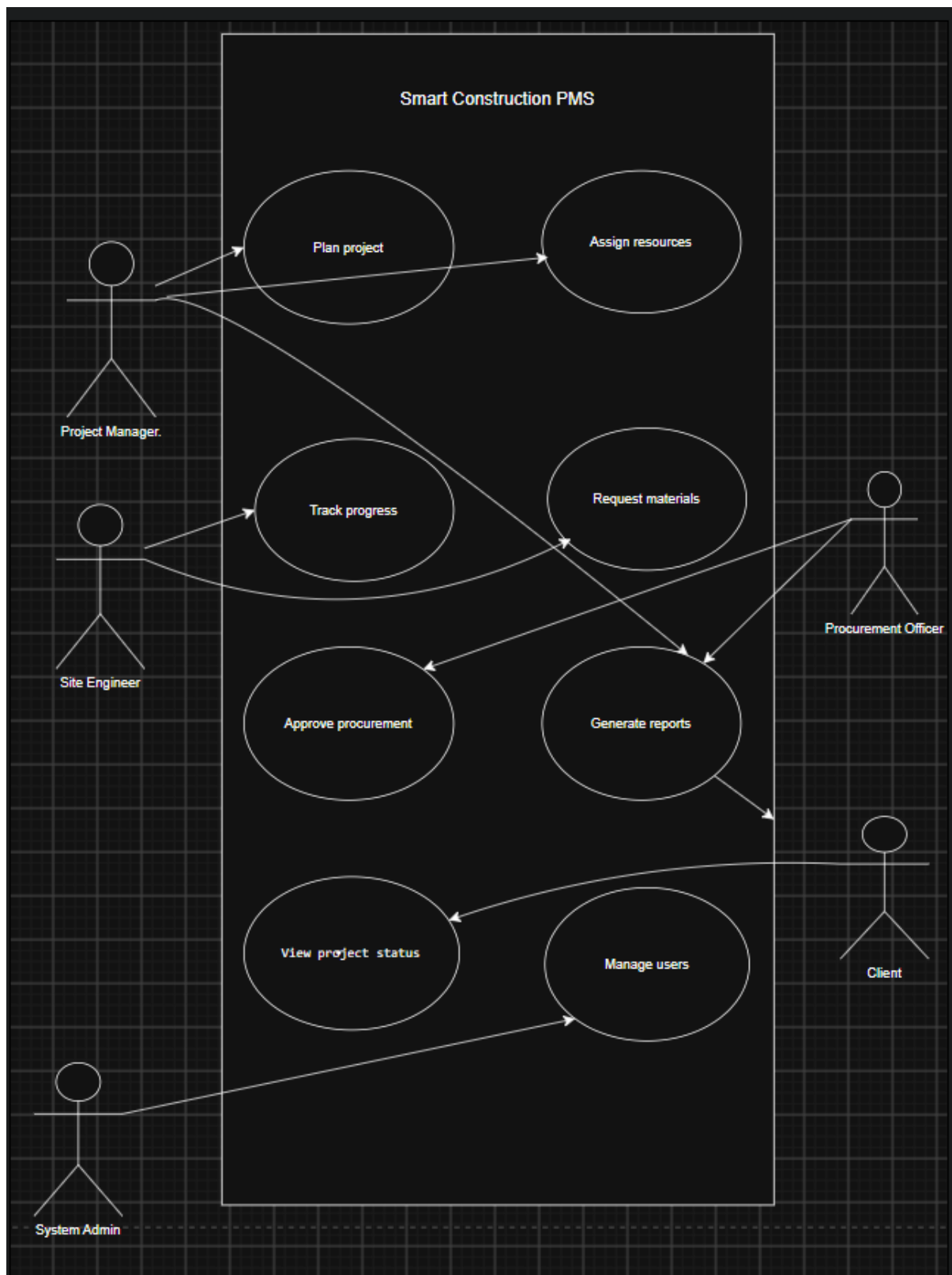
5.5 Software Engineering Principles Applied

- Modularity: Each function (procurement, scheduling, reporting) built as an independent module
- Scalability: Cloud-hosted architecture that scales with number of projects and users
- Reliability: Redundant backups and error handling to prevent data loss on-site
- Security: Role-based authentication and encrypted communication
- Maintainability: Clean separation of frontend, backend, and database layers
- Reusability: Common components (notification service, reporting engine) reused across modules

6. System Design

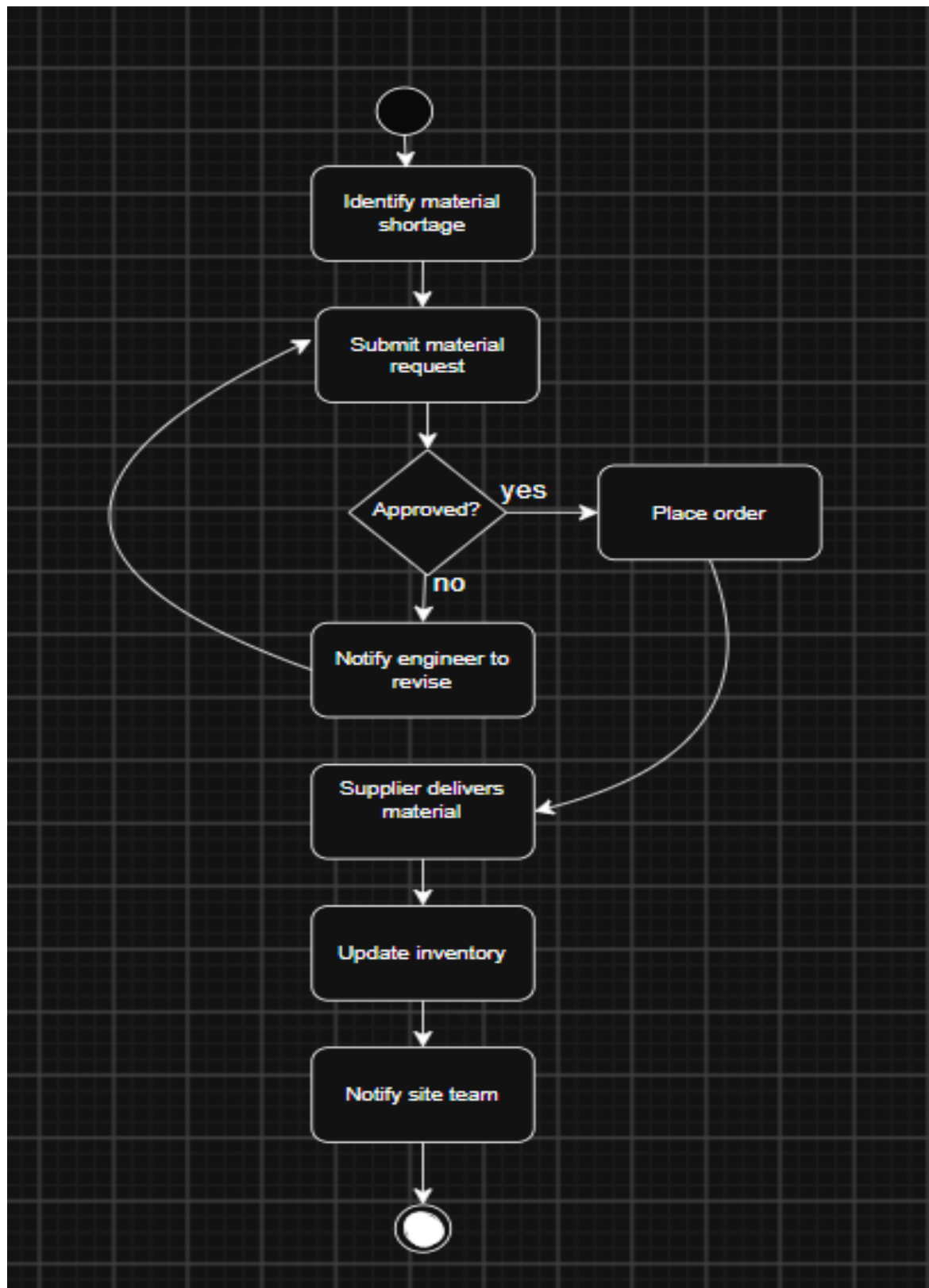
The following diagrams support this design and are included in the Appendix / attached separately:

6.1 Use Case Diagram



<https://drive.google.com/file/d/1cetlpOK10QzYifAN0A3DoKiZSBbj6mol/view?usp=sharing>

6.2 Activity Diagram (Material Procurement Workflow)



https://drive.google.com/file/d/11mrbWZbv61fMrtPwFPVgGK_ipQ926pEV/view?usp=sharing

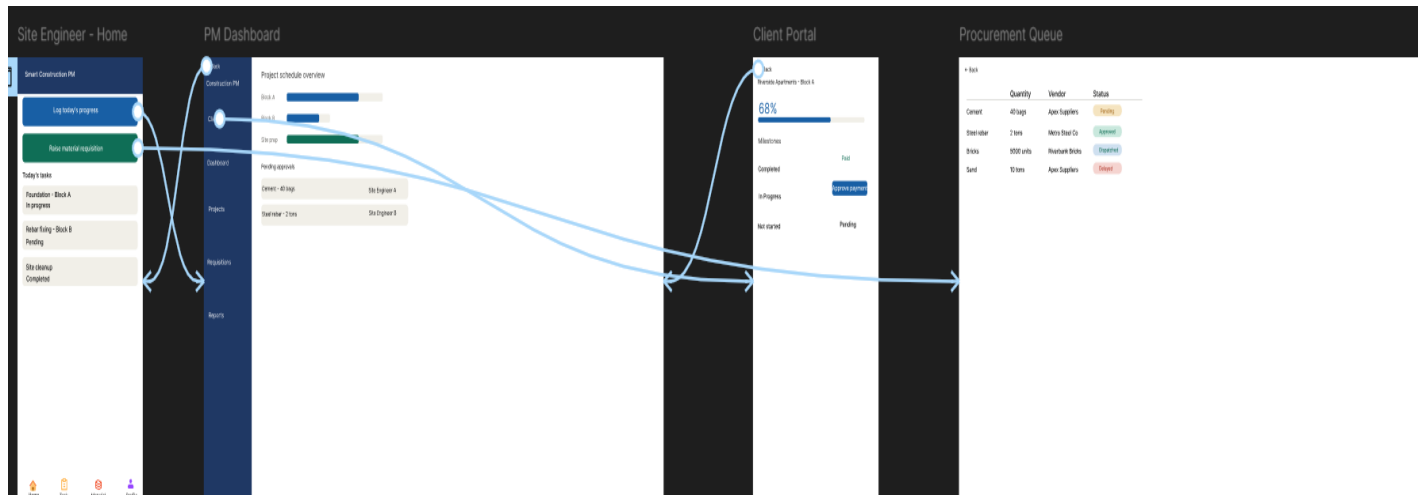
6.3 System Architecture Diagram



<https://drive.google.com/file/d/1dIYA3wFZacDclScemyHiIHOP2SIYm-8/view4?usp=sharing>

- Database ER Diagram
- Workflow Diagram (Project Progress Tracking)

6.4 Wireframes / UI Screens.



<https://www.figma.com/proto/fXSY8RtkemsJx9KifFnaoZ/Untitled?node-id=16-2&p=f&t=ugx417Yq8n7K6RKe-1&scaling=min-zoom&content-scaling=fixed&page-id=17%3A2&starting-point-node-id=16%3A2>

7. Methodology / SDLC Selection

Selected Model: Agile (Scrum)

Why this model:

Construction project management needs evolve as different stakeholders (site engineers, procurement, clients) provide feedback during development. Agile/Scrum allows the system to be built in short iterations (sprints), with each module (e.g., procurement, dashboard, notifications) delivered and tested incrementally. This reduces the risk of building a system that doesn't match real on-site workflows.

Advantages:

- Continuous stakeholder feedback from site engineers and project managers
- Faster delivery of usable modules (e.g., progress tracking can go live before the procurement module is complete)
- Easier to adapt to changing requirements as the company's project types vary

Suitability:

High, since requirements will evolve after real-world testing on live construction sites.

Development stages:

Sprint Planning → Design → Development → Testing → Sprint Review → Deployment (repeated per module)

8. Implementation Plan

Phase	Duration
Requirement Gathering	1 Week
Design	1 Week
Development	3 Weeks
Testing	1 Week
Deployment	1 Week

9. Expected Outcomes

- Benefits: Reduced material delays, better resource utilization, fewer cost overruns
- Business impact: Faster project delivery, improved client trust, lower operational costs
- User benefits: Site engineers save time on manual reporting; managers get real-time visibility
- Social impact: Improved infrastructure delivery timelines benefit local communities
- SDG contribution: Directly supports SDG 9 by modernizing infrastructure project delivery through innovation and digital tools

10. Risks and Limitations

- Internet dependency at remote construction sites
- Data privacy concerns around client and vendor information
- Security threats to centralized project data
- Budget limitations for smaller construction firms
- User adoption resistance from staff used to manual processes
- Technical challenges integrating with existing accounting / ERP systems

11. Future Enhancements

- AI-based delay prediction using historical project data
- Chatbot for quick site-level query resolution
- IoT sensors on equipment for automated usage tracking
- Predictive analytics for material demand forecasting
- Mobile app offline mode for low-connectivity sites
- Blockchain for tamper-proof procurement records

- Full cloud deployment with multi-region support

12. Conclusion

Construction companies face significant inefficiencies due to reliance on spreadsheets and manual communication, leading to delays and cost overruns. The proposed Smart Construction Project Management System addresses this by centralizing project planning, resource management, and progress monitoring into one platform. By applying Agile methodology and sound software engineering principles, the system can be delivered iteratively and adapted to real site conditions. This solution not only improves operational efficiency and profitability for construction firms but also contributes meaningfully to SDG 9 by promoting innovation in infrastructure delivery.

13. References

1. IEEE Xplore Digital Library – Software Engineering for Construction Management Systems
2. Springer – Digital Transformation in the Construction Industry
3. ACM Digital Library – Agile Methodologies in Enterprise Software Development
4. United Nations Sustainable Development Goals – Goal 9: Industry, Innovation and Infrastructure (sdgs.un.org)
5. World Bank – Infrastructure and Construction Sector Reports
6. Project Management Institute (PMI) – Construction Project Management Best Practices
7. Government e-Governance Standards for Infrastructure Projects